HARDIK BENIWAL

AI/ML Engineer
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PROFESSIONAL SUMMARY

AI/ML Engineer building production-grade Generative AI, RAG, NLP, and Computer Vision systems on AWS. MSc in Data Science & AI (IIT Madras & University of Birmingham). Skilled in LLM/VLM fine-tuning (PEFT/LoRA), scalable MLOps pipelines, and end-to-end ML lifecycle on SageMaker and Bedrock.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Bash
GenAI & LLMs: RAG, LangChain, LlamaIndex, AWS Bedrock, OpenAI & Anthropic APIs, Hugging Face Transformers, PEFT, LoRA, QLoRA, Prompt Engineering, Vision-Language Models (VLM), Fine-tuning, Evaluation
ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, CNNs, ViT, LSTM Autoencoders, Isolation Forest, Anomaly Detection, Time-Series, OpenCV
MLOps & Cloud: AWS (SageMaker, Bedrock, ECS, ECR, SQS, Lambda, S3, OpenSearch), Docker, Kubernetes, MLflow, DVC, FastAPI, Streamlit, Git, GitHub Actions, CI/CD
Data & Other: PostgreSQL, MySQL, MongoDB, OpenSearch (vector), FAISS, Tableau, Pennylane (Quantum ML), Responsible AI, Explainability

PROFESSIONAL EXPERIENCE

AI and Machine Learning Engineer | *Zenteiq* | Bengaluru, India **Aug 2026 – Present**

- Evaluated the **AhamX RAG pipeline** on quantitative and qualitative metrics; raised content-coverage accuracy from **~50% to >70%** by tuning the model's mode and parameters.
- Building the **agentic architecture of KogneX**, letting customers run scientific foundation models for inference and simulation.

AI/ML Engineer | *Resolve Tech Solutions (RTS)* | Bengaluru, India **Aug 2025 – Aug 2026**

- **Built an AI incident-management platform** linking OpenTelemetry/CloudWatch to ServiceNow; a confidence engine recommends or auto-creates tickets, enriched by multi-agent RAG over **20K+ logs/SOPs**, cutting manual triage **~40%**.
- **Built an S3 security-remediation platform** (AWS Bedrock/Claude 3.5, boto3, Streamlit) with a 3-stage **classify → risk → remediate** pipeline over **8+ bucket signals** and human-in-the-loop guardrails limiting fixes to whitelisted, dry-run-safe actions.
- **Deployed Isolation Forest + LSTM-Autoencoder anomaly detection** over **2M+ telemetry events/day**, cutting false positives **~35%**.
- **Engineered unsupervised alert deduplication** (HDBSCAN, Node2Vec, GraphSAGE) suppressing **~84% of duplicate SAP monitoring alerts** across **11K+ events**; deployed on AWS EKS + SageMaker with FastAPI and GitLab CI/CD.
- Received the company-wide **Townhall Award** for production-grade AI delivery within the first 6 months.

GenAI Research Intern | *Bosch* | Bengaluru, India **Feb 2025 - Jul 2025**

- Fine-tuned Vision-Language Foundation Models for vehicular automation across encoder-only, decoder-only, layer-freezing, and PEFT (LoRA) setups on a **4x A100** multi-GPU rig.
- Benchmarked **5+** VLM variants on internal driving datasets; improved **BLEU by ~18%, ROUGE-F1 by ~14%, and METEOR by ~11%** over baseline while cutting fine-tuning compute **~60%** via PEFT.

AI Intern | *Green Aadhaar* | Remote **Jan 2024 - Mar 2024**

- Led a real-time face-recognition attendance system in **OpenCV** + deep learning, achieving **90%+ recognition accuracy**; also built a real-time waste-detection model and explored AI for carbon management.

Research Intern | *Centre for Responsible AI (CeRAI), IIT Madras* | Chennai, India **Dec 2023 - Jan 2024**

- Contributed to Meta's AI-ethics initiative; analysed FLAN architecture for explainability and authored *"Ensuring Ethical Implementation of LLMs in EdTech"*.

SELECTED PROJECTS

Quantum-Enhanced Kernelised Contextual Bandits (MSc Dissertation, UoB) - Designed quantum bandit algorithms in RKHS using Pennylane; benchmarked quantum vs classical kernels with improved regret bounds. Supervised by Dr. Sharu Jose.

Melanoma Detection System (HAM10000) - Benchmarked **10** deep-learning architectures on **5K+** dermoscopic images (7-class); deployed best model (**~92% accuracy**) as a Streamlit app.

EDUCATION

MSc, Data Science and Artificial Intelligence (Joint Degree) - *Indian Institute of Technology Madras & University of Birmingham, UK* - CGPA: 7.12/10 2023 - 2025

BTech, Computer Science and Engineering - *Shri Mata Vaishno Devi University* - CGPA: 7.89/10 2018 - 2022

PUBLICATIONS

"Role of AI in Revolutionising EdTech", *Dimensions of Innovation and Technology in Rising India*, ISBN 978-93-6135-475-5 (June 2024).